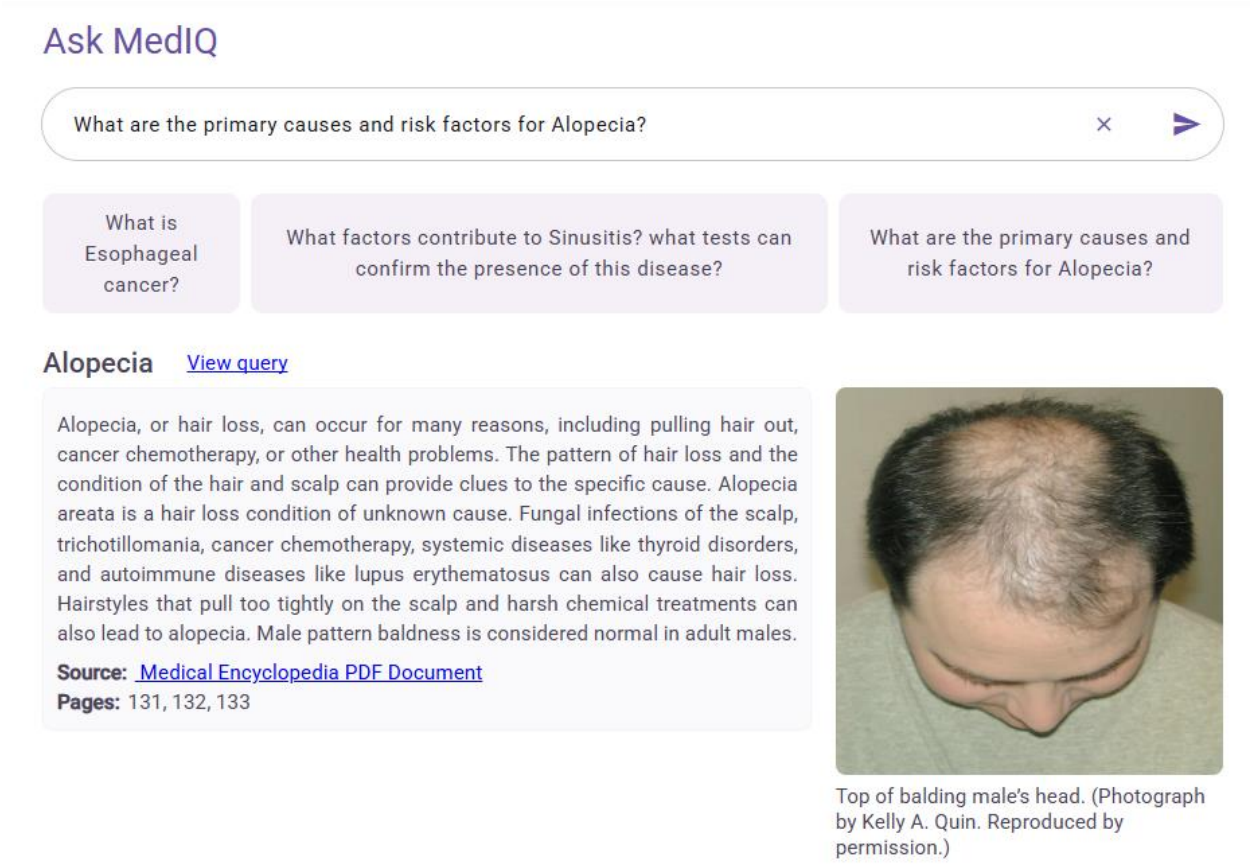
MedIQ Smart Medical Intelligence Platform User Guide

Goal:

Deliver **clinically scoped** responses that combine **disease knowledge** from the PDF medical encyclopedia with **diagnostic test intelligence** from the CSV, producing a unified **natural-language** answer that helps doctors, lab technicians, and clinical researchers make faster **point-of-care decisions**.

Example Query/Filters:

- What are the primary causes and risk factors for Alopecia?



Query

```
SELECT * FROM alloydb_usecase.search_medical_info('What are the primary causes and risk factors for
Alopecia?');

-- ==========================================================
--  FUNCTION: search_medical_info
--  PURPOSE:
--     Extract relevant medical info (chunks, tests, images)
--     and generate AI summaries based on user query.
--
--  RETURNS:
--     - disease_name
--     - summary_pdf (AI summary + pages)
--     - summary_csv (AI test summary)
--     - details_chunks[]
--     - tests_details[]
--     - related_images[]
-- ==========================================================

CREATE OR REPLACE FUNCTION alloydb_usecase.search_medical_info(
  user_query TEXT,
  embedding_distance_threshold FLOAT DEFAULT 0.3
)
RETURNS TABLE (
  disease_name   TEXT,
  summary_pdf    JSONB,
  summary_csv    JSONB,
  details_chunks JSONB[],
  tests_details  JSONB[],
  related_images JSONB[]
```

- What factors contribute to Sinusitis? what tests can confirm the presence of this disease?

What factors contribute to Sinusitis? what tests can confirm the presence of this disease?

What is Esophageal cancer?

What factors contribute to Sinusitis? what tests can confirm the presence of this disease?

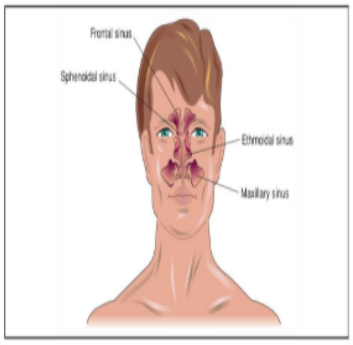
What are the primary causes and risk factors for Alopecia?

Sinusitis

View query

Sinusitis is an inflammation of the sinuses often caused by an infection. The sinuses are air pockets located within the bones of the face. When the lining of the sinuses is swollen, it interferes with the normal flow of mucus, trapping it and creating an environment for bacteria to grow. Sinusitis is almost always due to an infection, but allergies can mimic the symptoms. Bacteria like Streptococcus pneumoniae and Haemophilus influenzae are common causes. In children, Moraxella catarrhalis may be the cause. Fungi like Aspergillus, Candida, or Mucorales can cause sinusitis in people with weakened immune systems. Acute sinusitis usually follows an upper respiratory tract infection or cold, with symptoms like congestion, thick nasal drainage, fever, headache, pain over the sinuses, and sometimes jaw or teeth pain. Children may experience upset stomachs, and some patients develop a cough. Chronic sinusitis lasts for at least three months and rarely involves a fever. Symptoms include sinus pain and pressure, nasal congestion, sore throat, and bad breath due to drainage down the back of the throat.

Source: Medical Encyclopedia PDF Document
Pages: 3414



Sinusitis is the inflammation of the sinuses caused by a bacterial infection. Sometimes diagnosis may be problematic because the symptoms often mimic those of the common cold. Sinusitis is usually treated with antibiotics. (Illustration by Electronic Illustrators Group.)

Here are some diagnostic tests that can help confirm the presence of Sinusitis

1. White Blood Cells
2. Neutrophils
3. Lymphocytes
4. Monocytes
5. Eosinophils
6. Basophils
7. Immunoglobulin G
8. Immunoglobulin A
9. C-Reactive Protein

24. Base Excess
25. Calculated Total CO2
26. Oxygen Saturation
27. Ventilation Rate
28. Tidal Volume
29. PEEP
30. Oxygen
31. Intubated
32. Assist/Control
33. O2 Flow
34. Temperature
35. Bacteria
36. Leukocytes
37. Nitrite
38. RBC
39. Epithelial Cells
40. Yeast
41. Blood
42. Protein
43. Ketone
44. Bilirubin
45. Urobilinogen
46. Urine Color
47. Urine Appearance
48. Specific Gravity
49. Urine Mucous
50. Hyaline Casts

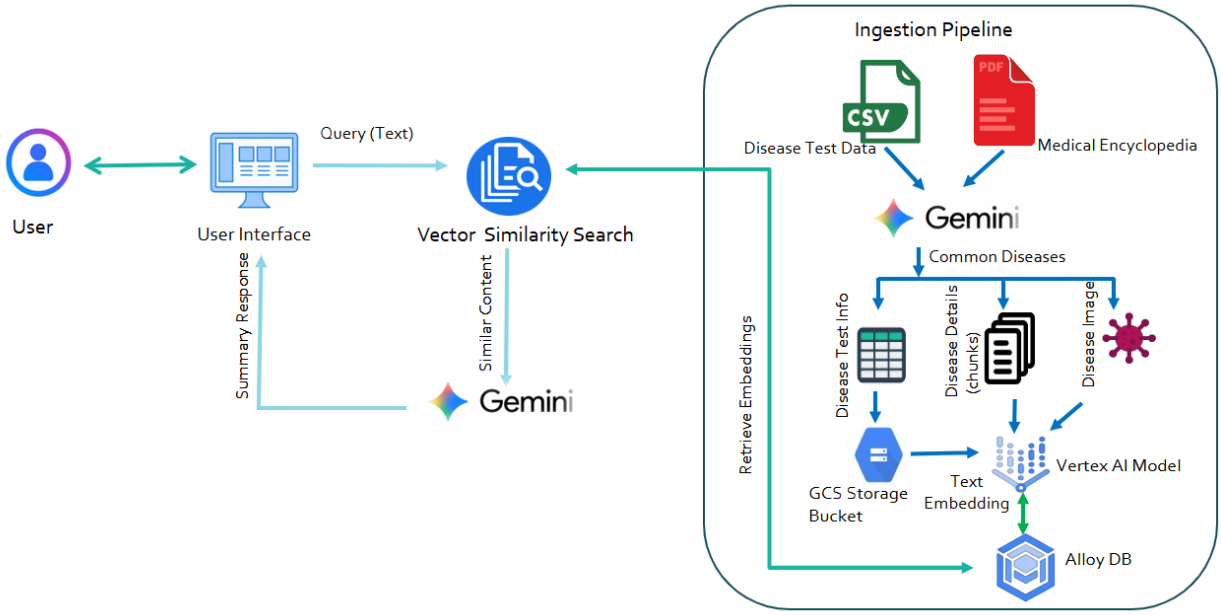
Source: Disease Test Confirmation CSV File

Key Alloy DB Components:

- **AlloyDB for PostgreSQL** (provides the database foundation)
- **Embeddings** creation (with google_ml extension) using text-embedding-005 model
- **Vector similarity** (with pgvector extension) using cosine similarity
- AlloyDB AI Query Operator: **ai.generate** for creating natural language response from search results
- **ScaNN** (optional, for large-scale deployments): Prefer ScaNN index with post-creation ANALYZE when working with very large corpora and if supported in your environment

Technical Challenge & Business Impact:

Challenge:
Coordinating **multi-source** (PDF & CSV sources) grounding directly inside SQL combining embeddings, **semantic retrieval**, and **response generation** across two distinct data sources combining power of unstructured PDF data and structured tabulated data (PDF chunks and CSV rows) within a single **TVF** (Table-Valued Function: A function that returns a table in PostgreSQL allows users to encapsulate logic in a reusable manner while outputting a set of records) demands careful query planning and orchestration.

Impact:
This demo produces higher-quality **clinical answers** exactly when needed, giving doctors and lab technicians a concise, **evidence-tagged** response that unifies **disease knowledge** with **diagnostic test checklists**, reducing the time spent cross-verifying multiple documents.

Architecture Diagram